

Optimality conditions: unconstrained problems

Sufficient optimality conditions

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Optimization: principles and algorithms



ÉCOLE POLYTECHNIQUE
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Sufficient optimality conditions

*The sufficient conditions are very similar to the necessary conditions.
Like the previous video, these are reminders of classical results from analysis.*

Sufficient optimality conditions

Theorem 5.7

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$$\nabla^2 f(x^*) > 0 \quad [\text{positive definite}]$$

Sufficient optimality conditions

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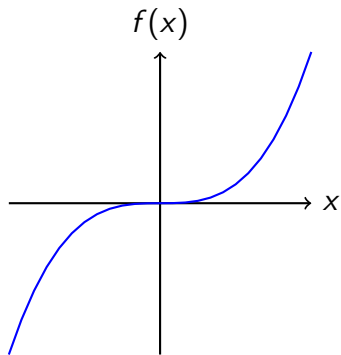
- ▶ and

$$\nabla^2 f(x^*) > 0 \quad [\text{positive definite}]$$

- ▶ then x^* is a local minimum of f .

Examples

$$f(x) = x^3, x^* = 0$$

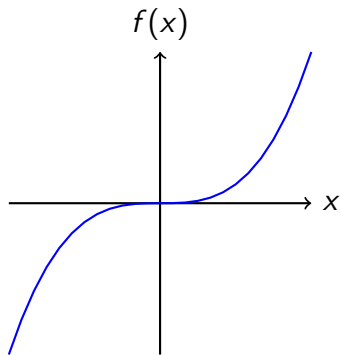


$$f'(x) = 3x^2, f''(x) = 6x$$

$$f'(0) = 0, f''(0) \neq 0$$

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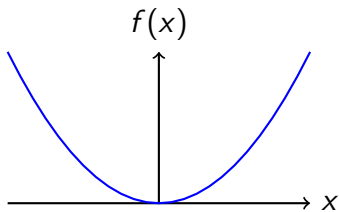
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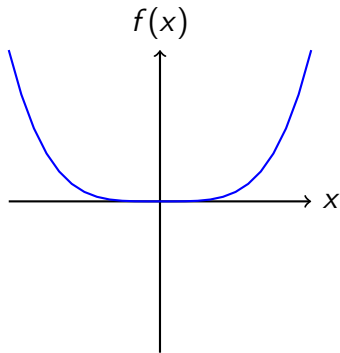


$$f'(x) = 2x, f''(x) = 2$$

$$f'(0) = 0, f''(0) > 0$$

Examples

$$f(x) = x^4, x^* = 0$$



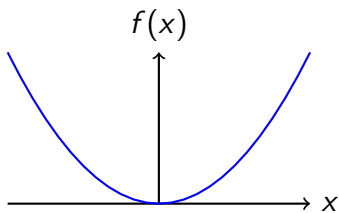
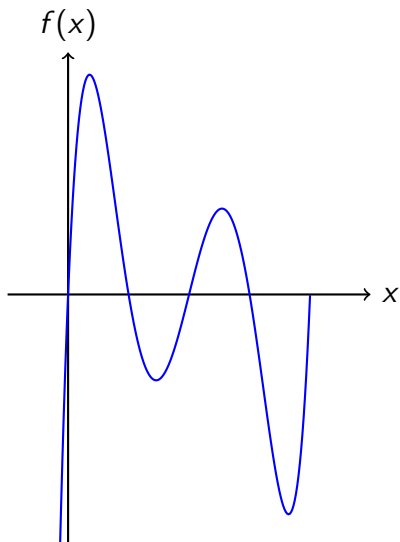
$$f'(x) = 4x^3, f''(x) = 12x^2$$

$$f'(0) = 0, f''(0) \neq 0$$

Global and local optimum

*Show that the sufficient conditions are verified both for local and global minima
Introduce the idea that a convex function has a global optimum.*

Global and local optimum



Global optimum

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- ▶ If f is strictly convex, x^* is the unique global minimum of f .

Global optimum

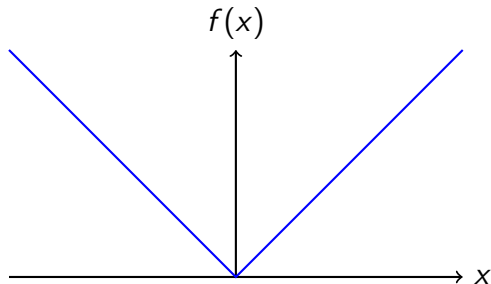
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Summary

*If the gradient is zero and the second derivative matrix is positive definite (all eigenvalues are strictly positive),
then the function is locally convex and we have a local minimum.
If the function is globally convex, a local optimum is also a global optimum.*